



ZETDC GIS PROJECT

USER'S WEB MANUAL

Version 1



Revision Sheet

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GIS USER'S MANUAL

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1.0 GENERAL INFORMATION

GENERAL INFORMATION

1.1 System Overview

The Geographical Information System (GIS) is set to bring the most spatially accurate and up-to-date electrical power infrastructure database available with accurate attribute data on a scale never before possible.

The purpose of the system is for:

- Planning, comparative analysis, project feasibility analysis and risk evaluation.
- Examining the geographic relationships among the various data components to identify new opportunities.
- Incorporating additional layers and proprietary information to increase analytical value.
- Creating a powerful visual presentations for project reporting.

GIS BENEFITS

The system is a web-based application that is set to bring the following benefits:

- i. Quick updates of network data
 - Accurate asset database
 - Reduced accidents
- ii. Quick retrieval of information, Information not centralised
- iii. Updated Base maps
- iv. Network represented up to meter point.
- v. Accurate asset register – realistic tariff
- vi. Efficient and optimised planning and designs, accurate project costs

2.0 SYSTEM SUMMARY

2.0 SYSTEM SUMMARY

2.1 System Configuration

The GIS Web Application is accessed by users using their devices- computers, laptops or cellphones connected on the Local Area Network (LAN). The application is accessible using any web browser of choice preferably Google Chrome.

2.2 User Access Levels

There are two different kinds of users in GIS depending on their permissions within the system:

- GIS Users- those with permissions to view and query data in the GIS web application
- Admin Users- those who have higher priviledges in the control of GIS at the back-end

3.0 USING GIS WEB APPLICATION

USING THE GIS WEB APPLICATION

To access the system, in your preferred web browser type in the GIS server IP address if known, or add the following address on the address bar: <http://zetdc.gis/>.

NB: The current GIS server IP address is 172.20.0.32

3.1 Settings

Users need to add exceptions to Internet options to be able to access GIS from the web browser. To add exceptions,

- In the Control Panel, follow path, Network and Internet → 'Network and Sharing Centre' → Internet Options
- In the Internet Properties dialog box, select Connections → LAN settings → Advanced
- In Proxy Settings dialog box, under Exceptions add the GIS server IP address, if known or add the link zetdc.gis

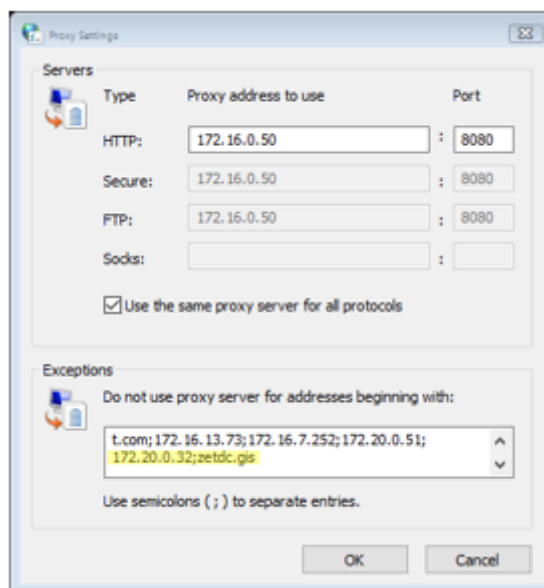


Figure Error! No text of specified style in document..1: Add exception for GIS

server

3.1.1 User Profile

User Profile is created by the administrator at request. To request for the creation of a new user:

- On the Log in page, select NEW ACCOUNT and the User Profile Request form appears

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Log in to continue.

ACCOUNT

EC Number Designation
Enter your ec number... Enter your designation...

Select centre ▼

First Name Surname
Enter your first name... Enter your surname...

E-mail Address
Enter your e-mail address...

LOG IN SEND REQUEST

Fig 3.2: New Account request

- Enter EC Number and designation
- Select center from the dropdown list
- Enter first name, surname and e-mail and select SEND REQUEST
- The system's administrator will approve and create the new account

3.1.2 Username

-The username is the user's ec number- ze + employee number ie. ze12345

3.1.3 Password

The password that corresponds to the current user is initially created by the administrator and the user will be able to change to a preferable password.

3.2 Getting Started

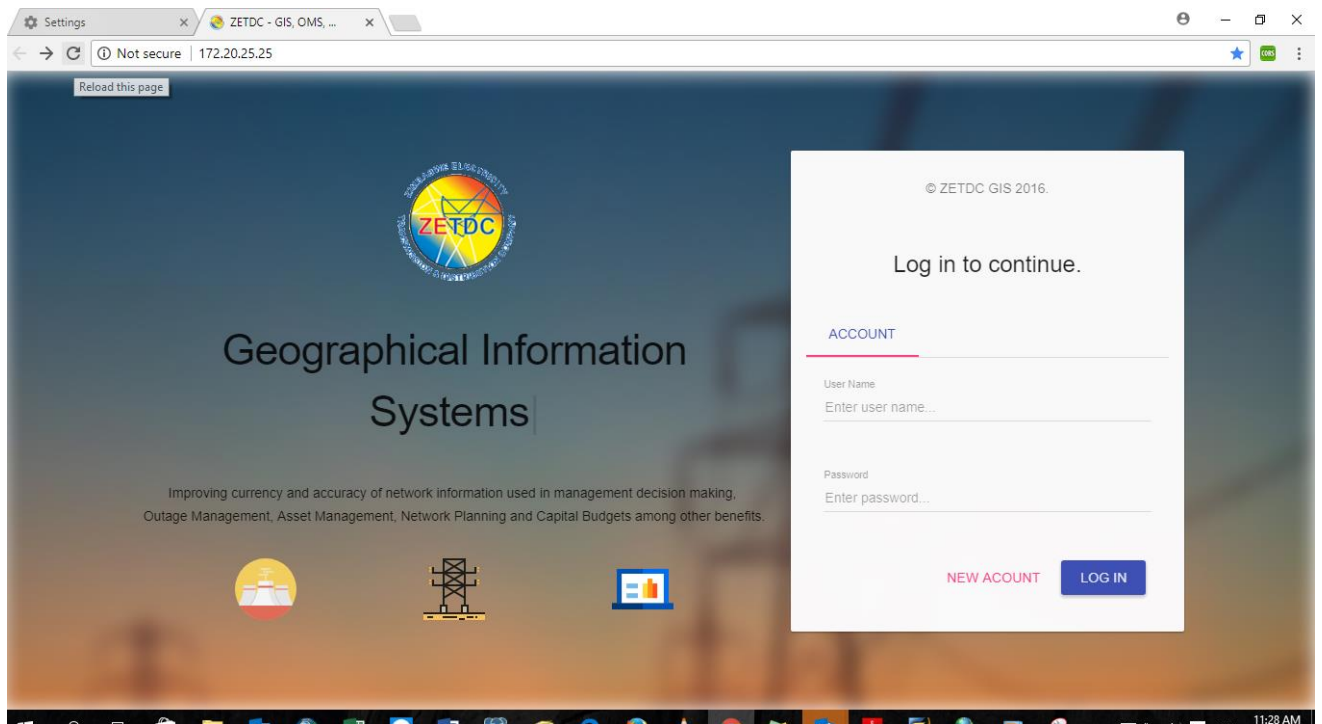


Fig 3.3: Home Page

3.2.1 Signing In

From the Home Page, enter username and password and select LOG IN

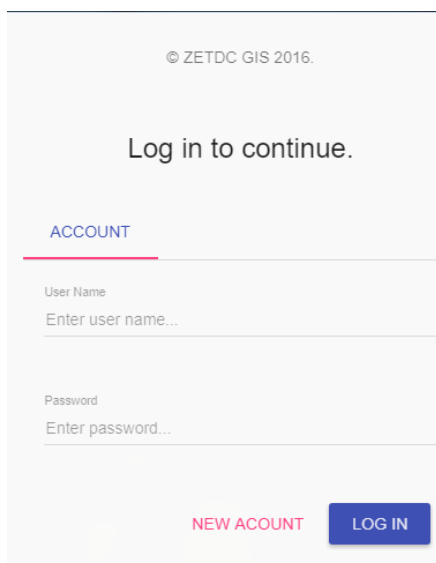


Fig 3.4: Log in

3.2.2.1 Toolbar

The toolbar contains the following icons:



Fig 3.7: ToolBar

Main Menu

The Main Menu icon on the Tool bar opens the Menu shown below:



Fig 3.8: Main Menu

The Main Menu contains list of:

- Home – Loads the Home Page from which ever point navigated to, within the application
- Map Legend-When the check box is selected, the Map Legend appears with Transmission and Distribution panels. To remove map legend from the screen,

deselect the map legend icon from the Main Menu. A map legend, is an explanatory list of symbols and features used within the map

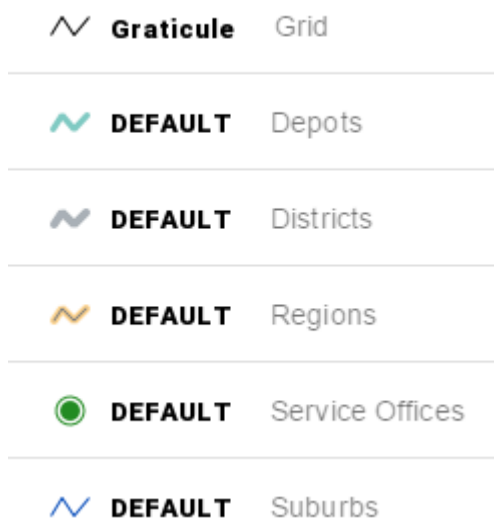


Fig 3.9: Map Legend

- Engineering Reports- The reports showing for Engineering are the Network Asset Registers and Network Change Logs

Engineering Reports

- Select the Main Menu icon 
- Select Engineering Report and the GIS Engineering Reports page opens

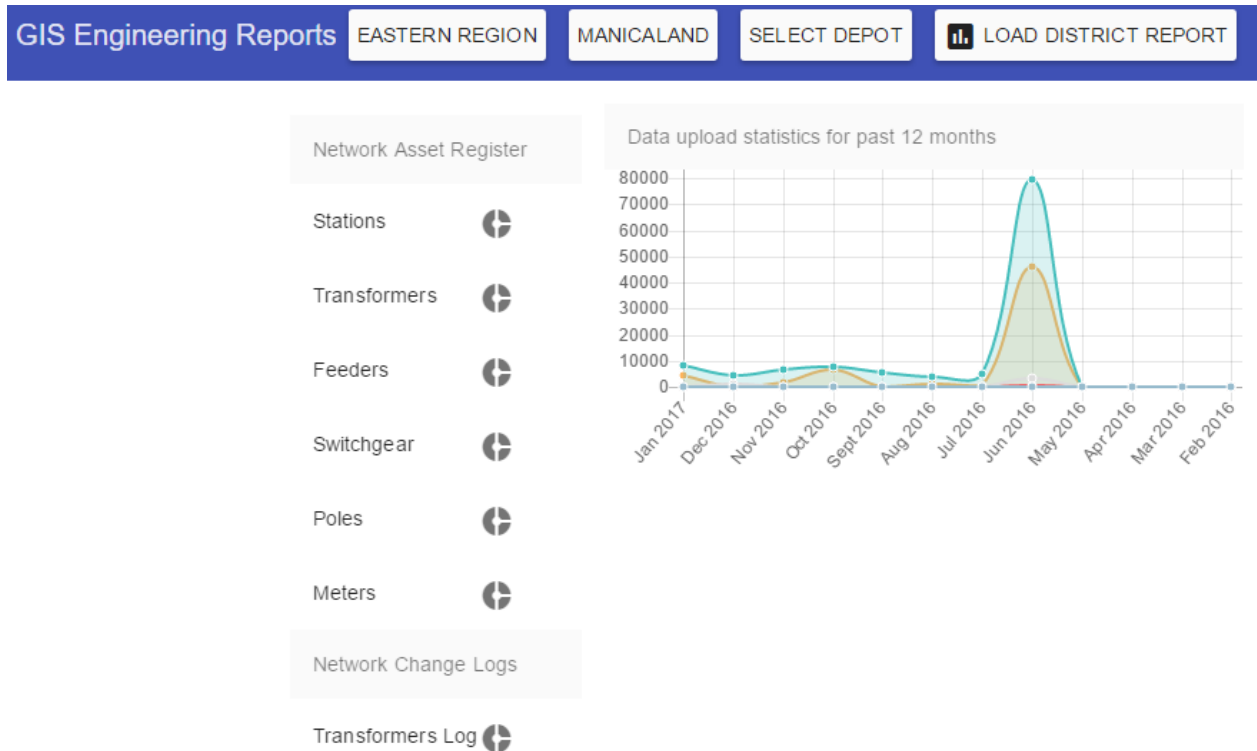
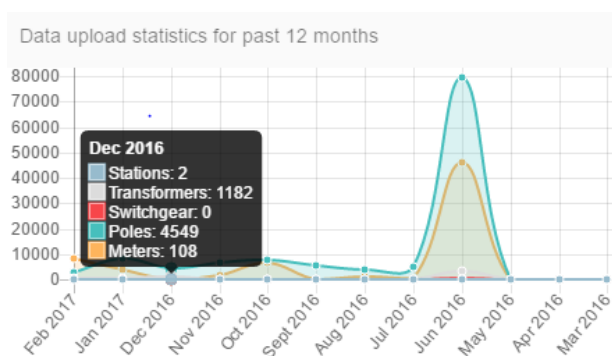


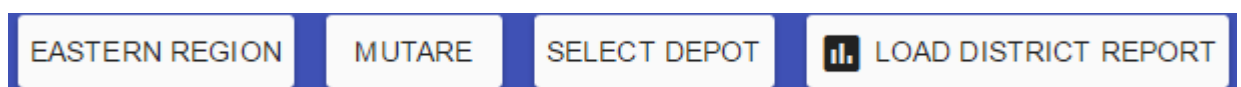
Fig 3.10: Engineering Reports

The Data upload graph shows the monthly uploaded feature data from all regions. Select the period to view the uploaded document



To view Reports:

- Select the report to be viewed, from the Network Asset Register, for example Stations
- Select region, district or depot depending on which report is to be viewed

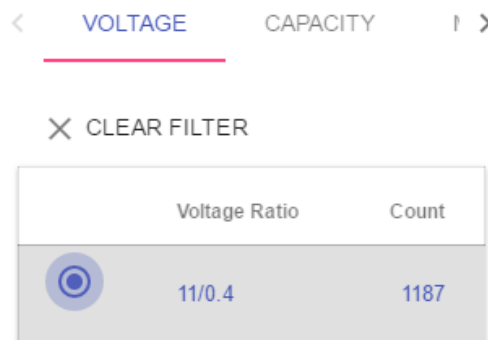


- Select Load Report

- There are Save and Share options



- Share with gives the option to share the report with another user
 - Save as Excel allows the user to save and export as an Excel file
 - Save as Doc allows the user to save and export as a Word document
 - Save as PDF allows the user to save and export as a KML file
 - Save icon allows the user to export the document as an Excel and save it for later viewing through the browser
- The Register menu on the left pane of the screen shows filtering options with attributes depending on the asset selected
 - To filter, select the attribute to filter by, for example on Transformers one can choose to filter by voltage, select the condition to filter by, for instance the voltage ratio



- To remove filter condition, select CLEAR FILTER

NB: To use the Search, Selection Tool and Identifying feature on the web application:

- Select the layer you want to work with, for instance select Transformers on the Properties Panel
- Select the tool you want to use, for instance Search, Selection Tool or Identifying Feature
- If a layer is not selected, these features will be greyed out and cannot be used as shown in the figure below

Reset

When using tools such as the measuring tool, to clear or reset the current tool select the reset button. The reset button cancels any tool that is currently being used.

Dashboard

This option is a short cut to the dashboard, which includes links to GIS, EIS, Mobile Jobs, Settings, Admin.

Measuring Area Tool

The Measuring Area tool is used to measure the area selected by the user. To use this tool:

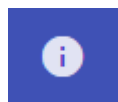
- Click on the 'Start Measuring Area' icon 
- Take note of the Properties Panel which will contain the Measure Area data and the Options toast at the bottom right corner of the page
- On the Options Toast, the following options are available:
 - Measure Area without aggregation- measures area of different polygons without collective total showing on the Properties Panel
 - Measure Area with aggregation- measures area of different polygons with collective total showing on the Properties Panel
 - Reset measurements- Clears measurements showing on the Properties Panel
- Select the option you want to use to measure the area, a blue dot appears on the Data Area or Map View Area
- Click around the area you want to measure and double click on the last point and the measured area is displayed on the Properties Panel
- Multiple polygons can be measured without reloading the page
- To clear the measurements of area on the Properties Panel, select the Reset Measurements on the Options Toast

Measuring Distance

The Measuring Distance tool is used to measure length selected by the user. To use this tool:



- Click on the 'Start Measuring Distance' icon
- Take note of the Properties Panel which will contain the Measure Length data and the Options toast at the bottom right corner of the page
- On the Options Toast, the following options are available:
 - Measure line without aggregation- measures length of different distances without collective total showing on the Properties Panel
 - Measure line with aggregation- measures length of different distances with collective total showing on the Properties Panel
 - Reset measurements- Clears measurements showing on the Properties Panel
- Select the option you want to use to measure the length, a blue dot appears on the Data Area or Map View Area
- Click the length you want to measure and double click on the last point and the measured length is displayed on the Properties Panel
- Multiple lines can be measured without reloading the page
- To clear the measurements of length on the Properties Panel, select the Reset Measurements on the Options Toast



Identify Feature

The Identify Feature tool is used to select and give details of a particular feature within a specific layer. To use the Identify Feature tool:

- Select the layer in which you want to identify the feature, for example Transformer

Note: The transformer and switchgear layers shows all features on the map as points



- Select the Identify Feature tool on the Main Toolbar
- For easier navigation, zoom in on the area with the transformer to be identified

- Select the item to be identified in this case, transformer

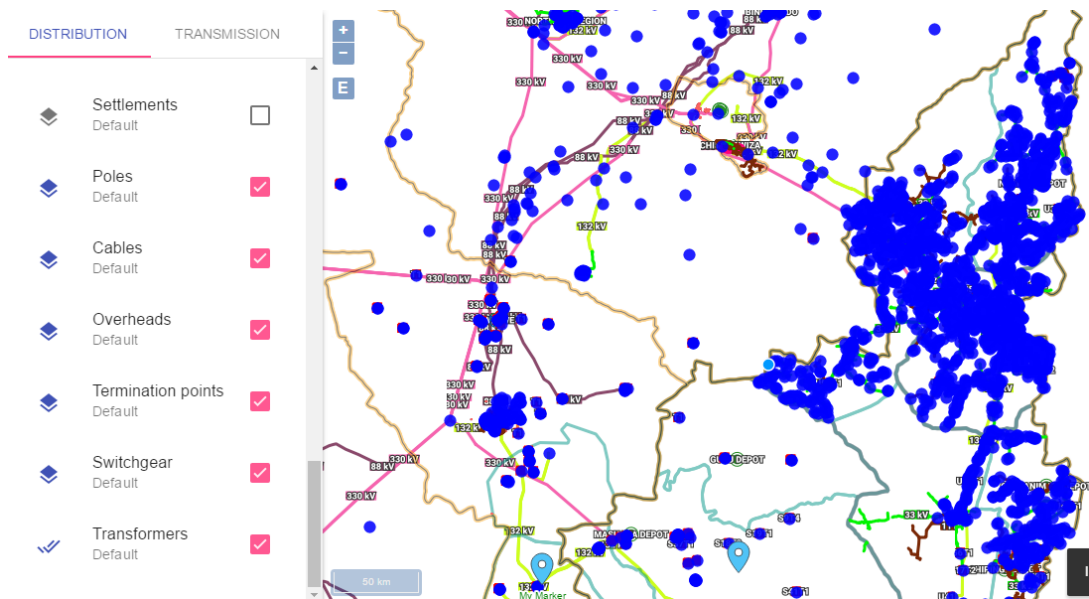


Fig 3.11: Identify feature on Transformer layer

- Note, the Properties Panel will show all captured details of selected feature, in this case transformer and the Options Toast will show the position of the identified feature with options to switch from the identified feature to default layer on the Properties Panel.
- The NEARBY tab, when selected displays features in the nearby area in which the user would have selected and the details are as shown in the figure below

TRANSFORMER

NEARBY

Transformers Nearby

s290 11/0.4 11/0.4
No system number



s20 11/0.4 11/0.4
No system number



- The FILES tab allows the user to upload files pertaining to that specific feature for example a transformer or switchgear. File types supported include word documents, videos, excel sheets, pdfs, images. To upload a file, drag and drop selected file from its location to the section labelled '*Drop files here to upload*' or click on the labelled section and the Open dialog box appears, browse to the necessary file and upload.

Fly To

'The Fly to' feature allows quicker navigation to a particular point. To zoom in on a particular point, one can either:

- Double click on specific area until the required point is reached
- Use the '+' feature on the Map View 
- Fly to the same point 

To fly to:

- Select the 'Fly To' tool on the Main Toolbar
- On the Data Frame or Map View Area, click on the area you need to fly to

To zoom out:



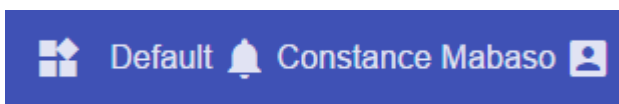
- On the Data Frame or Map View Area, use the '-' feature to zoom out or use "E" to load page to its original state

Scale



The map scale shows the relationship (or ratio) between distance on the map and the corresponding distance on the ground. By zooming in and out of the map, the scale will change automatically. However, to change the scale manually, select scale on the toolbar and select the desired ratio. The scale above shows that every 1mm on the screen represents 10000 mm on the ground.

Main Toolbar ...contd



The right end of the Main Toolbar shows the user's name, apps menu icon and account menu icon.

Map Thematics



Map Thematic icon gives a list of all map themes available.

- Thematic Maps- A thematic map is a map that emphasizes a particular theme or special topic such as accident zone or network status.
 - Navigate to and select Thematic map

- Select the map theme you would like to use from the eight options: Default, Accident zone, Maintenance, Statistical, Outages, Network status, Proposed network and Network Changes.
- Changes are noted on the Data Frame

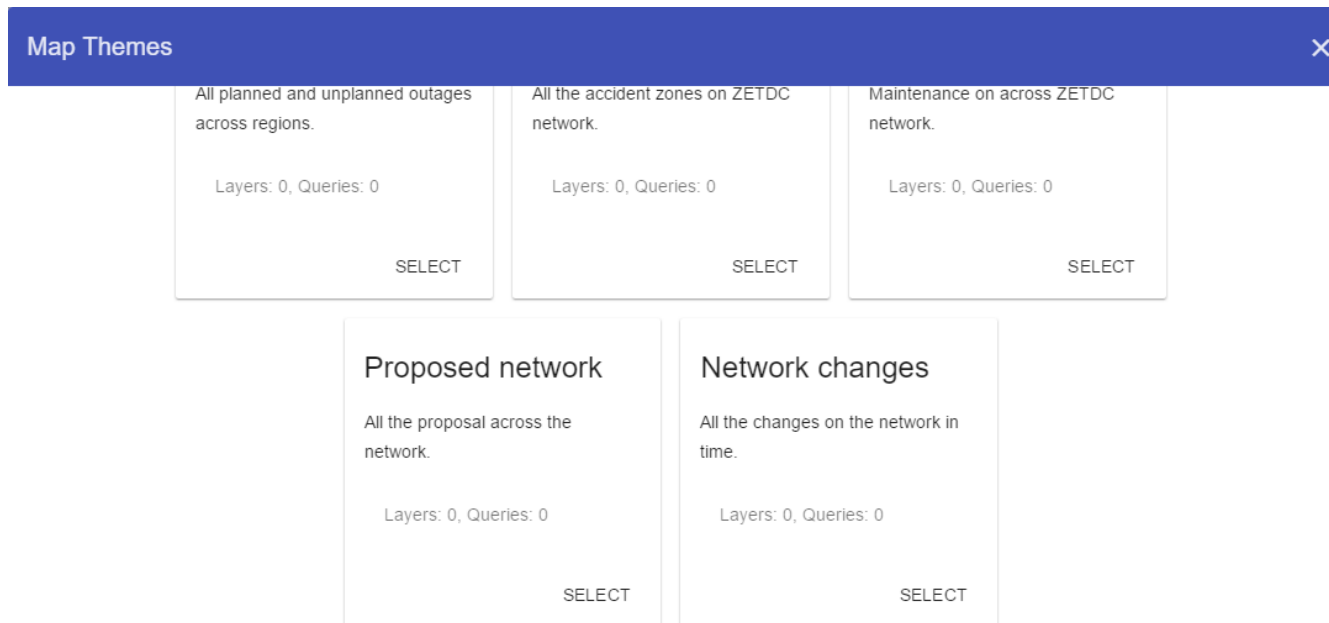


Fig 3.12: Map Thematics

Account Menu

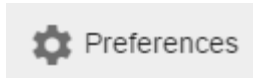
The Account Menu allows the user to access their account profile. To access Account Menu:

- Select the Account Menu icon 
- A pop-up list appears with:

Preferences

Preferences are settings that allows the user to make preferred changes to the GIS as opposed to default settings. To access Preferences:

- From the Account Menu icon, select the Preference item in Account Menu list



- From the Preferences page, select the Theme to configure by using the drop down i.e. Defaults, Network Status, Statistical etc. Layers in the Theme selected appear with their default settings
- For each layer there is a switch button which is grey when turned off and purple when turned on, Select on the button to either switch on or off preference
- Select the Save button at the top of the page

The Default button on the Preferences page will restore all preferences to their default settings.

Sign Out

To sign out of the GIS and return to the Sign In page:

- From the Account Menu Icon, select Sign Out

Search

Search Icon is used to search features within the map. To use the search option:

- Navigate to the Properties Panel on the left of the web page
- Select the feature item you would like to search, for example, Transformers and the search panel appears



Note-The More Option shows that a layer has been selected and the single tick on the right shows the Transformers are visible on the map, if deselected all transformers will not show on the map. Select the More Option to Edit Overlay Symbology or to export the selected layer to KML.

- On the Search window, one can optionally specify region, district, or service center.

Specify the search criteria by:

- Select the condition either IS or NOT
- Select the attributes of the feature for example, transformer has attributes such as name, system number, voltage
- Select comparison: Equals- searching for a result which exactly matches the parameter

Like- the result is similar, but not exactly identical to the parameter

Greater than, Greater or Equal to, Less than, Less or equal to-comparisons used for numeric values

- On parameter, enter the value to be searched for and select enter

Fig 3.12: Advanced Search on transformer layer

Note: On using the LIKE comparison, use % sign before, in between or after the parameter in this way-

- %tare – searching for anything that includes **tare** at the end but starts with any value e.g. mutare, matare, old mutare
- %tare% - searching for anything that includes **tare** in between e.g. mutare 2, old mutare
- Muta% - searching for anything that includes **muta** at the beginning e.g. mutare, mutambara, mutare 2

3.2.2 Layers

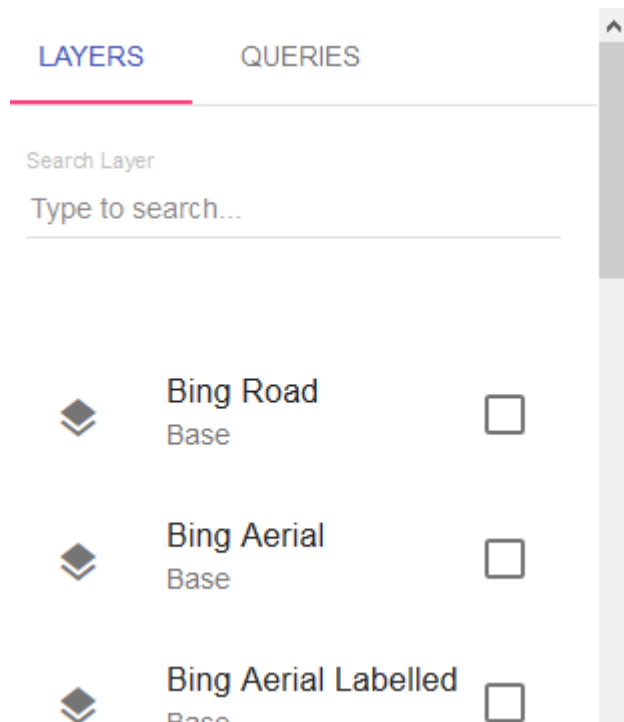
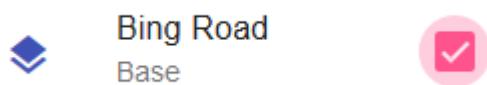


Fig 3.13: Layers Panel

The properties panel contains Layers and Queries options. The Layers option shows different layers that the users can select and view on the Map view. Layers include:

- Base maps - Layers labelled base are base maps that load background imagery of different maps when selected. To load base maps, select checkbox at the right of the desired map



NB-Base maps use online services therefore are affected by internet connections.

- Boundaries – There are layers that show different boundaries such as political, residential and ZETDC boundaries such as wards, suburbs and depots.
- Features – There are layers that show features (assets) such as poles, transformers and switchgears and by default they are selected on the properties panel

To search a layer, click on the 'Type here' text box and type in layer to be searched.

Search Layer

Type to search...

3.2.3 Queries

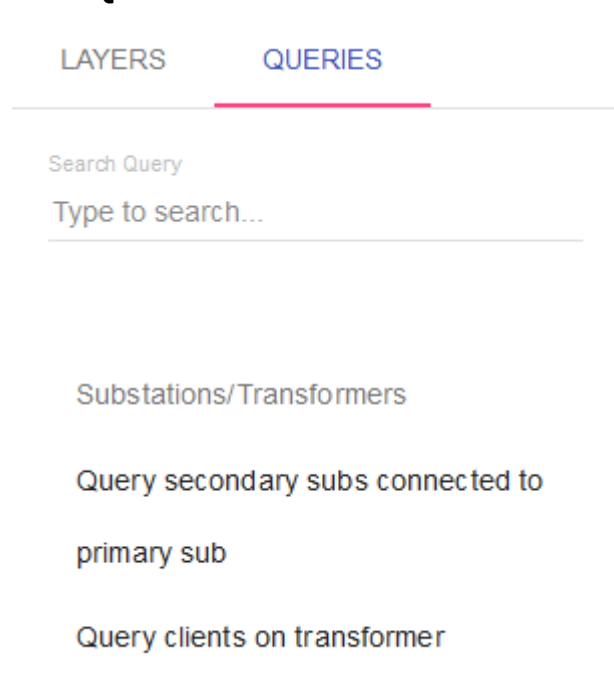


Fig 3.14: Queries Panel

Queries option on the properties option gives a list of predefined queries one can use to manipulate data. Queries are on substations/transformers and clients/meters.

To search for a query, type in text to be searched and the search function will filter the result.

To use queries

- Select query to use, for instance Query secondary subs connected to primary sub
- Select the primary substation you would like to query for secondary substations by using the identifying tool by selecting 'USE IDENTIFY TOOL'
- Select the transformer on the map view

- Select 'USE CURRENTLY IDENTIFIED TRANSFORMER', transformer name is shown on the pop up window

Query Secondary Subs Connected To Primary Sub

BASIC

CLEARWATER U45 T2 33/11 U45T2

Please select the primary substation you would like to query for secondary substations

Step 1 USE IDENTIFY TOOL ⓘ **USE CURRENTLY IDENTIFIED TRANSFORMER**

Stopping Conditions

HELP CANCEL GENERATE

Fig 3.14: Query on Secondary Subs

- Select Stopping conditions, on Normally open points select status either Ignore or Stop at Open Points and on Switchgear status select status Ignore or Stop at open switches
- Select 'GENERATE'

The query results are displayed on the left, on the map view as all secondary substations connected to the selected primary substation and on the right of the screen and appear as below:

Connected Secondary Subs To Primary Sub



SUMMARY

DATA

Statistics

Voltage Ratio	Subs	Capacity (kVA)	Subs
33/0.4	306	50.0	377
33/0.23	4	100.0	208
11/0.23	52	25.0	272

Summary shows:

- Total of secondary substations connected to a primary substation against the voltage ratio and against their capacity

Data lists all the secondary substations connected to the selected primary substation.

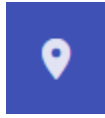


The above tools on the result panel are used for the following:

- Zoom- zoom to result extent
- Share- Share query with someone
- Export to Excel- Export queried results as a Microsoft Excel document
- Export to Word- Export queried results as a Microsoft Word document
- Export to PDF- Export queried results as a PDF document
- Save as Excel- Save queried results as Excel document

3.2.4 Markers

Markers are indicators that one uses to mark out points of interest. To mark a point of interest:



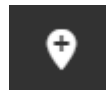
- Select the marker icon on the Main Toolbar
- The markers toast will appear at the bottom right of the screen



- To add a new marker with known coordinates select this icon
- Capture details of the marker and select SAVE



- To add a marker with unknown coordinates, select the add new marker icon
- Click on the point of interest which you would like to mark on the map
- A pop up window as shown below appears, add the marker name , description of marker and type of marker and select SAVE



New Marker

Latitude	Longitude
-19.303795691468764	30.4415597912457

Name of Marker
Enter name of marker

Description
Enter description of marker

0 / 150

Type of Marker

CANCEL SAVE

- To view markers, go to the Markers tab on the tab strip. A list of all markers will appear.

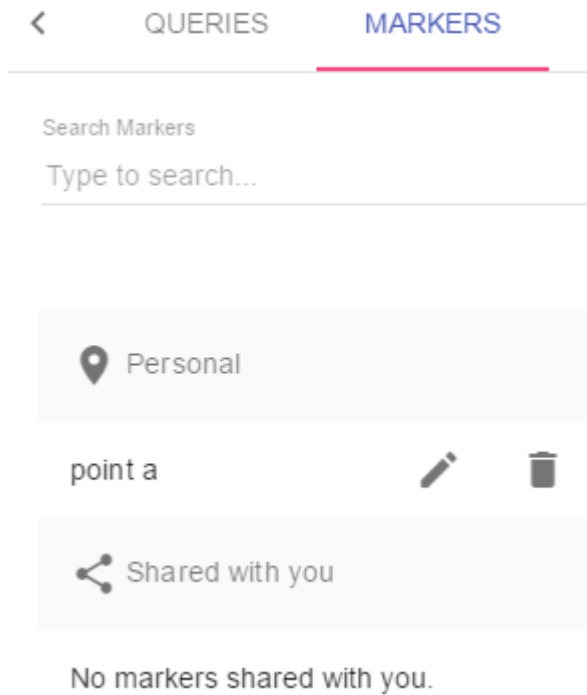


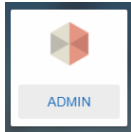
Fig 3.15: Markers Panel

- To edit the marker name select the edit option  and save
- To delete the marker, select the delete option
- To move the marker from one point to the other, select the marker from the list, select the marker icon on the Main Toolbar and select the edit marker on the marker toast . Select on the marker to be moved and drag it to the new location. Select anyway on the map to save the new location

4.0 ADMINISTRATION

4.0 ADMINISTRATION

To log in to the Administration Profile, select the ADMIN icon on the dashboard



Note: Only users with administrative privileges can access the Admin profile.

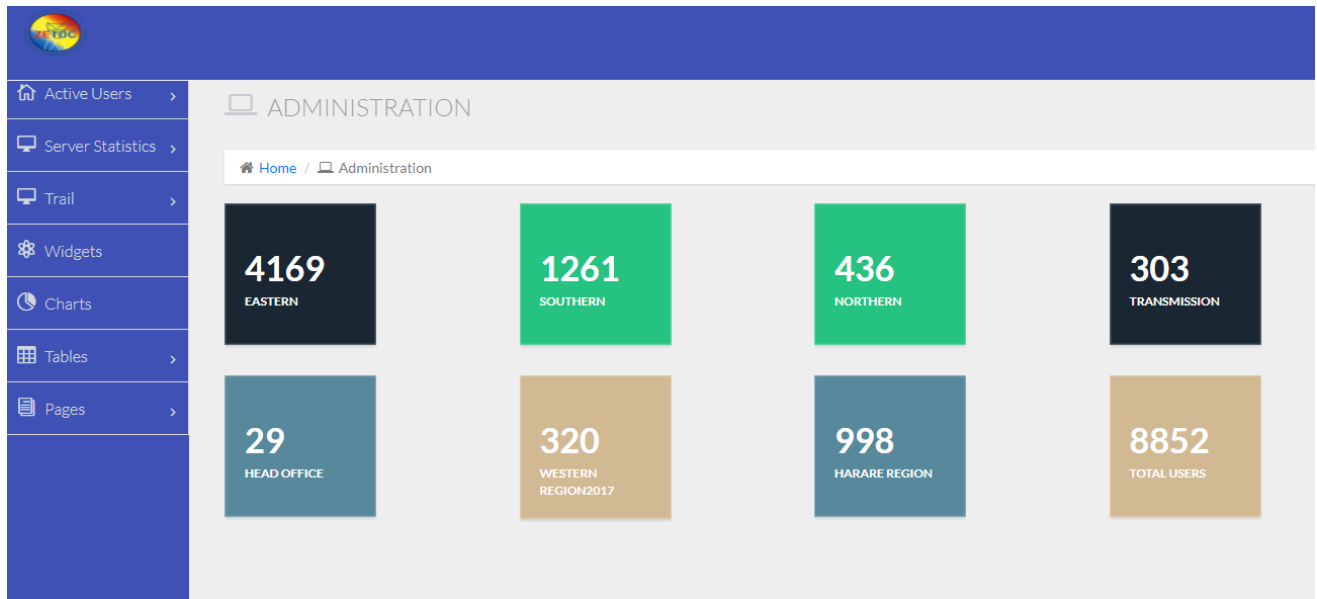


Fig 4.1: Admin Landing Page

Administration landing page shows counter statistics of all regions and head office log in sessions to date.

4.1 Active Users

4.1.1 New Users

Username *	
Centre *	
First Name *	
Surname *	
Designation *	
Password *	
confirm Password *	
Email *	
Save	

Fig 4.2: User Form

To create new users from requests made from system:

- Select NEW USERS
- From create new user form, select 'CREATE FROM EXISTING', if the user requested for a profile through the system
- From the list of users who requested profiles, select the user whose profile is to be created
- The information of the user is automatically filled in the form, if any corrections, edit and save, and the user is created

To create new users without user request submitted:

- Select NEW USERS
- Enter all fields of creating a new user
- On Username, use EC number
- On Centre, select from the drop down list the center the user is stationed
- All fields are mandatory to fill in, first name, surname, designation, password and select save

4.1.2 Edit User

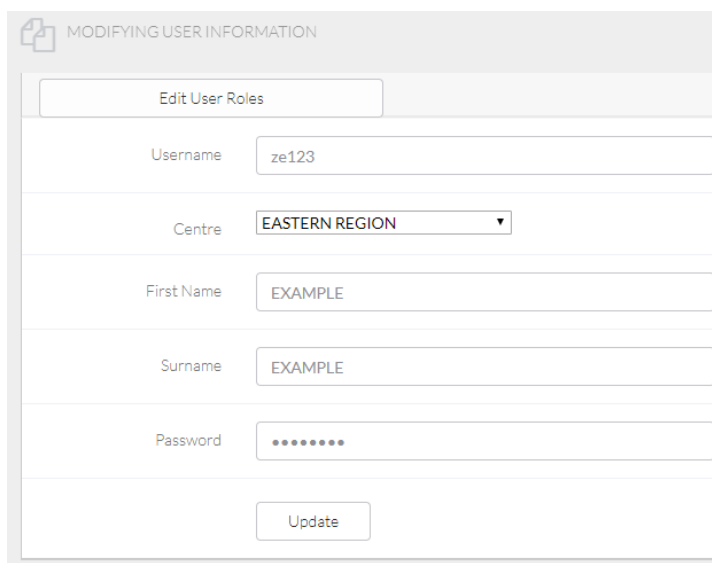


Fig 4.3: Edit User Form

- To edit a user, select 'Edit User', under Active Users
- To select name enter the username of the user, i.e. ze123 and select Submit
- Edit user's details and select Update to save changes made

4.1.3 Edit User Roles

- Select Edit User under Active Users
- Enter username ie ze268313 and Submit
- Select 'Edit User Roles'
- The username and roles already assigned to the user if any appear

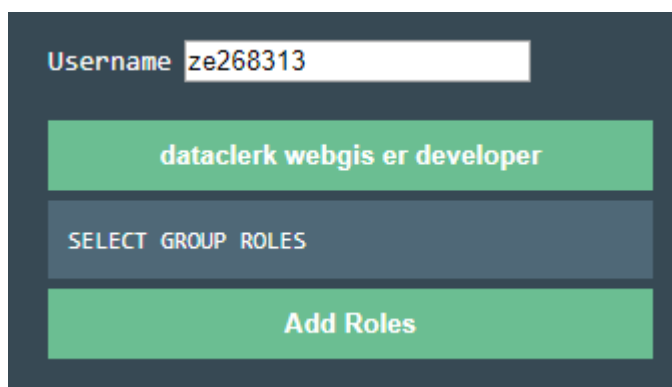


Fig 4.4: Shows User ze268313, with roles dataclerk, webgis. Er and developer

- To add roles or assign roles from this screen, select group roles to be assigned
- Select checkboxes with corresponding group role(s)
- Select 'Add Roles'

4.1.4 Group Roles

Roles are permissions defined for the system, which are assigned to users by the administrator

Group Role	Definition of Role
Admin	Overall permissions for GIS administrator user interface and database
dataclerk	Insert, update, delete data on user's specified schema
datagis	Overall permissions for gis schema
developer	Overall developer permissions
sr	User is zoomed into Southern Region in GIS user interface and has permissions to insert, update and delete data in sr schema
hr	User is zoomed to Harare Region in GIS user interface and has permissions to insert, update and delete data in hr schema
nr	User is zoomed to Northern Region in GIS user interface and has permissions to insert, update and delete data in nr schema
er	User is zoomed to Eastern Region in GIS user interface and has permissions to insert, update and delete data in er schema
tr	Insert, update, delete data in transmission schema
treast	User is zoomed to Transmission East in GIS user interface and has permissions to insert, update and delete data in treast schema
trwest	User is zoomed to Transmission West in GIS user interface and has permissions to insert, update and delete data in trwest schema
webgis	Allows user to log in to the GIS user interface

4.2 Server Stats

4.2.1 Mobile Statistics

Mobile Statistics shows the usage of mobile GIS

- To view Mobile Statistics, go to Server Stats and select Mobile Statistics
- Select Centre of interest from the drop down list
- Select the period in which data was collected and Submit

- The report populated will show all users with mobile GIS profiles in the requested center and the populated data captured on features including meters, transformers, switchgears and poles.

Home / Table / GIS Mobile Data Collection Report For / MUTARE URBAN DEPOT / Period: / 2017-11-10 to 2017-11-14

Username submit PDF representation Excel representation

District	Centre	Username	Artisan	Meters	Transformers	Switchgears	Poles
MUTARE DISTRICT	MUTARE URBAN DEPOT	ze265292	STANLEY MADZIMBA	0	0	4	0
MUTARE DISTRICT	MUTARE URBAN DEPOT	ze265292	MUSHEENI MANDISWANA	0	0	0	0
MUTARE DISTRICT	MUTARE URBAN DEPOT	ze265292	JOSEPH MUKOMBE	35	0	0	0
MUTARE DISTRICT	MUTARE URBAN DEPOT	ze265292	JOHN ZVIZAI	0	0	0	0
MUTARE DISTRICT	MUTARE URBAN DEPOT	ze265292	MARIAN MURUDZAI	0	0	0	3

Fig 4.5: Mobile Stats

- Using this report, one can filter out data by the username, first name or surname
- Report can be exported to PDF or Excel document

To view further details of features updated by the user,

- Select on the username (Artisan) you would want to view
- Select feature to view-ie Meters, Transformers etc

172.20.0.10 / localhost / 172.20.0.32 / home/admin / 172.20.0.32 / home/admin/mobileusage/meters.php?id=meters

Meter Number	Depot	Tariff	Make	MCB size	Customer Name	Bill Address	Username	Suburb	Date Synced	Action Done	Comments
1620022119	Mutare Urban	Domestic	Ittron	60A	Mangezi	3726 Hobhouse	ze265292	Hobhouse	2018-04-13 08:35:49	insert	
30710045540	Mutare Urban	Commercial	Iskra	60A	Netone	Zimta Booster	ze265292	Zimta	2018-01-22 16:56:55	insert	New connection
30710045532	Mutare Urban	Commercial	Iskra	60A	Netone	Sakubva stadium	ze265292	Sakubva	2018-01-22 16:57:35	insert	
30710044287	Mutare Urban	Commercial	Iskra		Netone	Afm Greenside	ze265292	Greenside	2018-01-22 16:56:56	insert	
3173052381	Urban	Domestic	Inhemeter	60A	Mubaiwa Kudzai Mwakutuya	11368 greenside ext	ze265292	Greenside	2018-01-22 16:56:56	insert	

Fig 4.6: Mobile Stats Detail Report

Appendix A

Acronyms and Abbreviations

GIS – Geographical Information System

LAN – Local Area Network

PDF – Portable Document Format

KML - file format used to display geographic data in an Earth browser such as Google Earth

END

A decorative graphic consisting of three horizontal lines. The top line is red, the middle line is yellow, and the bottom line is blue. They are all of the same length and are positioned one above the other.